

Task03 Theory Answers

1. `int.Parse` throws exception on null, `Convert.ToInt32(null)` returns 0.
2. `TryParse` avoids exceptions and improves user experience.
3. `GetHashCode` helps hashing objects in dictionaries and hash sets.
4. Reference equality means two variables point to same object instance.
5. `String` is immutable for safety, performance optimizations, and thread safety.
6. `StringBuilder` uses mutable buffer to reduce new allocations.
7. `StringBuilder` is faster for repeated large modifications.
8. Interpolation is most used because it is readable and clean.

Part02

1. `String` immutability means every change creates a new string object.
2. `Enum` is a named set of constants. Used for statuses, options, categories.

Common enums: `DayOfWeek`, `ConsoleColor`, `StringComparison`.

3. Use `string` for fixed/light text. Use `StringBuilder` for many edits.

Bonus

4. Self study report: practiced parsing, references, strings, `StringBuilder`, enums, arrays, linked lists.
5. User defined constructor is a constructor written by developer to initialize object state.

6. Array vs Linked List:

Array fixed size, fast indexing, contiguous memory.

Linked List dynamic size, easy insert/delete, slower indexing.